

Sai Naga

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Summary

- Seasoned data professional with deep hands-on experience designing and operationalizing production-grade **ETL/ELT and streaming** pipelines across **AWS** and **Azure**, focused on scalable, reliable data platforms.
- Adept at **Python** and **SQL** engineering on **Snowflake**, **Redshift**, and **BigQuery**, orchestrating transformations with **Apache Spark**, **Airflow**, and **dbt** that hold up under real production load.
- Proficient in **dimensional modeling**, **data warehousing**, and **medallion (Bronze/Silver/Gold)** architectures, delivering trusted, well-modeled datasets that power **Power BI** and **Tableau** analytics.
- Experienced in building **AI-powered** data layers spanning **RAG** pipelines, **embeddings**, **vector search**, and **LLM** integration via **Snowflake Cortex**, **AWS Bedrock**, and **LangChain**.
- Skilled in applied **machine learning** including **predictive modeling**, **fraud/risk scoring**, and **feature engineering**, with production **MLOps/ModelOps** using **scikit-learn** and **PyTorch**.
- Grounded in **statistical analysis**, **A/B experimentation**, and **exploratory data analysis (EDA)**, surfacing trends and actionable insights from large, complex datasets to drive business decisions.
- Strong command of **data governance**, **RBAC**, **data quality**, and **observability (SLAs/SLOs)**, containerizing with **Docker** and **Kubernetes** and automating delivery through **CI/CD**.

Skills

Languages:	Python, SQL, R, Scala, Bash
AWS Cloud:	S3, Glue, Redshift, EMR, Step Functions, Lambda, EC2, ECS/ECR, IAM, CloudWatch, CloudTrail, Lake Formation, Bedrock, SageMaker
Azure/GCP:	Azure ML, Azure OpenAI, Data Factory, Synapse; GCP BigQuery, Vertex AI
Data Eng:	Apache Spark/PySpark, Airflow, dbt (Cloud, Macros), Apache Hudi, Kafka, ETL/ELT, Medallion (Bronze/Silver/Gold)
Warehouses:	Snowflake (Cortex, Time Travel, Zero-Copy Cloning, RBAC), Redshift, BigQuery, Databricks (Delta Lake)
Databases:	PostgreSQL, MySQL, MongoDB, Apache Hive, Redis, Vector DBs (FAISS, Pinecone, Chroma)
AI/ML:	scikit-learn, PyTorch, pandas, NumPy, Regression, Classification, Clustering, Time Series, NLP, Feature Engineering, SHAP
GenAI/LLM:	RAG, LLM Fine-tuning (LoRA), Embeddings, Vector Search, Prompt Engineering, LangChain, OpenAI/Azure OpenAI, Claude API
BI & Viz:	Power BI, Tableau, Matplotlib, Seaborn, Plotly
DevOps:	Docker, Kubernetes, Terraform, CI/CD, Jenkins, GitHub Actions, Git
Practices:	Data Governance, Data Quality, Compliance, MLOps/ModelOps, AI-Assisted Dev (Copilot, Cursor), Agile/Scrum, Jira/Confluence

Experience

Data Engineer

Jan 2025 – Present

State Farm

Chicago, IL · Hybrid

- Operationalized production-grade **ELT pipelines** into a **medallion (Bronze/Silver/Gold)** lakehouse, expanding analytics-ready coverage by **40%** by modularizing **Python** ingestion across 15+ sources into **Snowflake**.
- Architected **RAG**-ready data layers for GenAI underwriting assistants, cutting research time from hours to minutes by generating **embeddings** served through **vector search** wired into **LLM** frameworks.
- Engineered **fraud-detection** and **risk-scoring** feature pipelines for **P&C** claims, lifting model precision by **18%** via standardized **Python/SQL** features and production **ModelOps** with data science.
- Embedded **LLM** capabilities in the data layer with **Snowflake Cortex** and **AWS Bedrock**, accelerating insight delivery by **30%** by automating classification and summarization of unstructured claims text.
- Built modular **dbt** models and macros across a multi-repo **dbt Cloud** project, cutting transformation defects by **25%** by enforcing **schema-evolution** standards, documentation, and automated tests.
- Instrumented **observability** with metrics, alerting, and **SLAs/SLOs** across 50+ pipelines, raising availability to **99.9%** via automated monitoring on **CloudWatch** and **CloudTrail**.
- Accelerated delivery by **35%** by adopting **AI-assisted development** tools (**GitHub Copilot**, **Cursor**, **Claude/OpenAI APIs**) to generate code and optimize heavy **Snowflake** queries.
- Hardened **data governance** with **RBAC** and compliance guardrails, eliminating unauthorized-access findings by securing environments through **IAM** and **secure CI/CD** workflows.

Data Engineer

Aug 2022 – Aug 2024

Alstom Transportation

Hybrid

- Architected scalable **ETL/ELT** pipelines on **AWS Glue**, **Step Functions**, and **Apache Hudi**, consolidating 20+ sources into an **S3** data lake and cutting ingestion runtime by **45%**.
- Scaled processing of **50M+** daily records by tuning **Apache Spark (PySpark)** through partitioning, caching, and **Parquet** compaction, lowering compute cost by **30%**.
- Owned metadata and schema discovery with **AWS Glue Data Catalog** and **Crawlers**, improving dataset discoverability by **40%** via standardized naming conventions and documentation.
- Modeled curated marts in **Amazon Redshift** with **star schemas** and tuned sort/dist keys, improving BI query performance by **50%** for analytics and reporting teams.
- Containerized workloads on **AWS ECS/ECR** and codified infrastructure with **Terraform**, shrinking new-pipeline setup from days to hours through reusable modules.
- Engineered real-time ingestion with **Kafka** and **Kinesis** streaming **100K+** events/min, enabling sub-second downstream availability via exactly-once processing.
- Established **secure CI/CD** with automated testing, version control, and code reviews, cutting production defects by **35%** by enforcing engineering standards across the team.
- Implemented monitoring, logging, and alerting on **CloudWatch**, cutting mean-time-to-detect failures by **40%** via SLA-backed health checks on critical pipelines.

Data Engineer

Jun 2019 – Aug 2022

Ernst and Young

On-Site

- Delivered **ETL/ELT** solutions across client engagements, raising reporting accuracy by **30%** by integrating disparate sources into governed warehouses with robust standardization.
- Built **Azure Data Factory** and **Synapse** pipelines in client environments, enabling scalable cloud processing of **10M+** records via dependency-aware orchestration.
- Wrote performant **SQL** across **PostgreSQL**, **MySQL**, and **Apache Hive** to reconcile large datasets, raising data-quality scores by **25%** through validated transformations.
- Designed **star-schema** dimensional models and curated layers, reducing ad-hoc query turnaround by **35%** by simplifying self-service reporting structures.
- Developed **Power BI** dashboards and reports, speeding stakeholder decisions by **40%** by translating raw data into clear, actionable visualizations.
- Built early **predictive models** (regression/classification) with **scikit-learn** and **pandas**, improving forecast accuracy by **15%** through targeted feature engineering.
- Automated recurring workflows with **Python** and **Airflow**, eliminating **20+ hours** of manual effort weekly via scheduled, validated pipeline runs.
- Partnered with cross-functional stakeholders to convert business rules into tested transformations, ensuring maintainability by documenting logic and lineage in **Git**.

Education

University of Illinois Chicago

Chicago, IL, USA

Master of Science in Computer Science

Certification

Microsoft Certified: Azure AI Engineer Associate

Feb 2026 – Feb 2027

Credential ID: 237E973316A0BAC1

Projects

GenAI Knowledge Assistant (RAG) | *Python, LangChain, FlowiseAI, Pinecone, FastAPI, Docker*

- Grounded an **LLM** in a private corpus via **embeddings** and **vector search**, returning source-cited answers.
- Orchestrated retrieval in **FlowiseAI** and containerized with **Docker**, cutting answer latency by **40%**.

Fraud Detection & Risk Modeling Platform | *Python, scikit-learn, Snowflake, SHAP, Airflow*

- Trained **classification** models on **Snowflake** features, flagging high-risk transactions at **0.92 AUC**.
- Automated retraining and **model monitoring** in **Airflow** with **SHAP** explainability and drift checks.

Cloud-Native Medallion Lakehouse | *AWS Glue, Apache Hudi, Redshift, dbt, Terraform*

- Built a **Bronze/Silver/Gold** lakehouse on **S3** with **Glue** and **Hudi** upserts, serving **Redshift** marts via **dbt**.
- Provisioned infrastructure with **Terraform** and automated tests and deploys through **CI/CD**.

Real-Time Streaming Analytics Pipeline | *Kafka, Spark Structured Streaming, Snowflake, Power BI*

- Streamed **Kafka** events through **Spark Structured Streaming** into **Snowflake** with sub-minute freshness.
- Surfaced live KPIs in **Power BI** with windowed, exactly-once aggregations for near real-time monitoring.